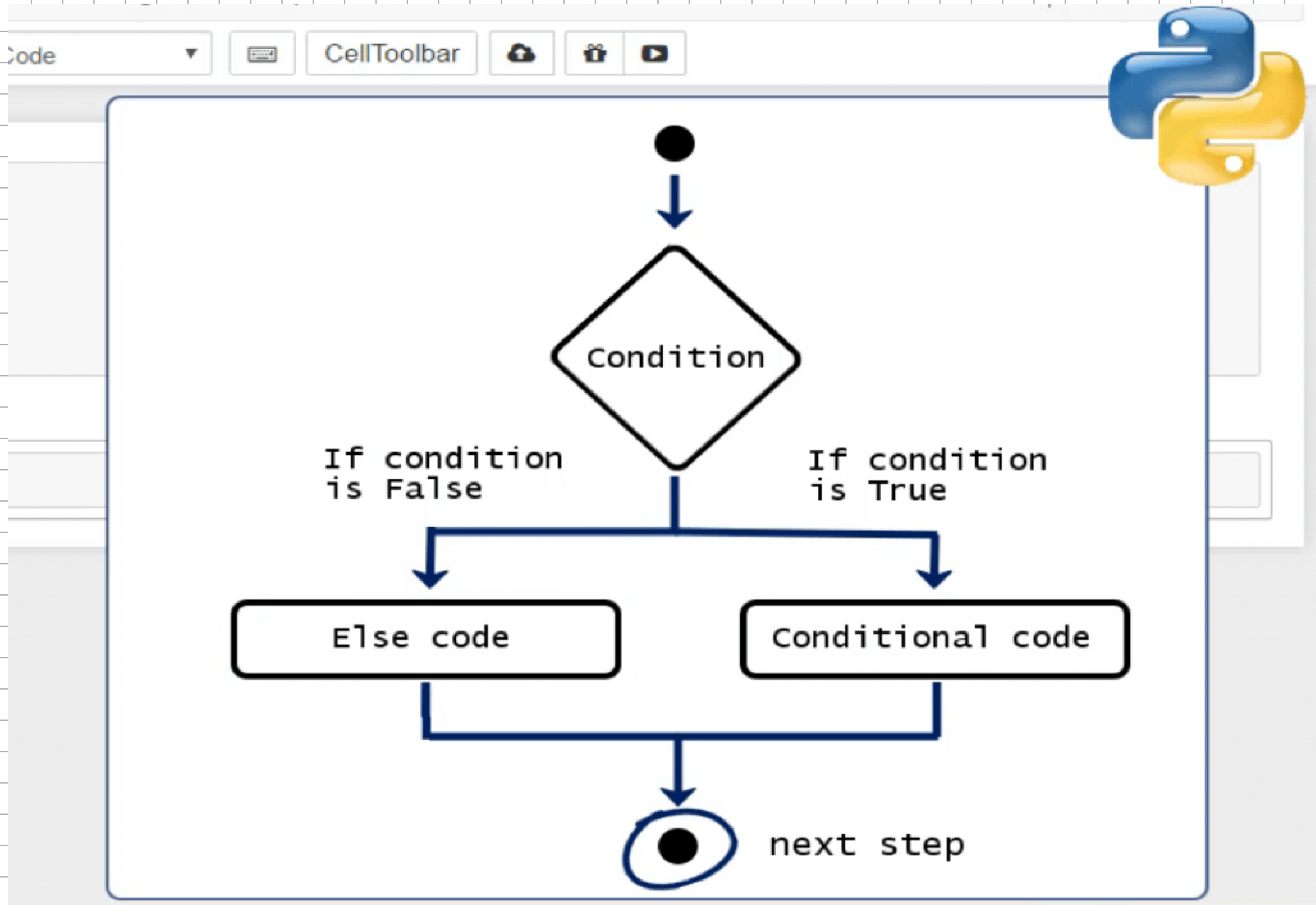


Outer if Condition Is True

```
score = 85
has_passed_exam = True

if score >= 60:

    if has_passed_exam:
        #code
    else:
        #code

else:
    #code

#code after if Else
```

Outer if Condition Is False

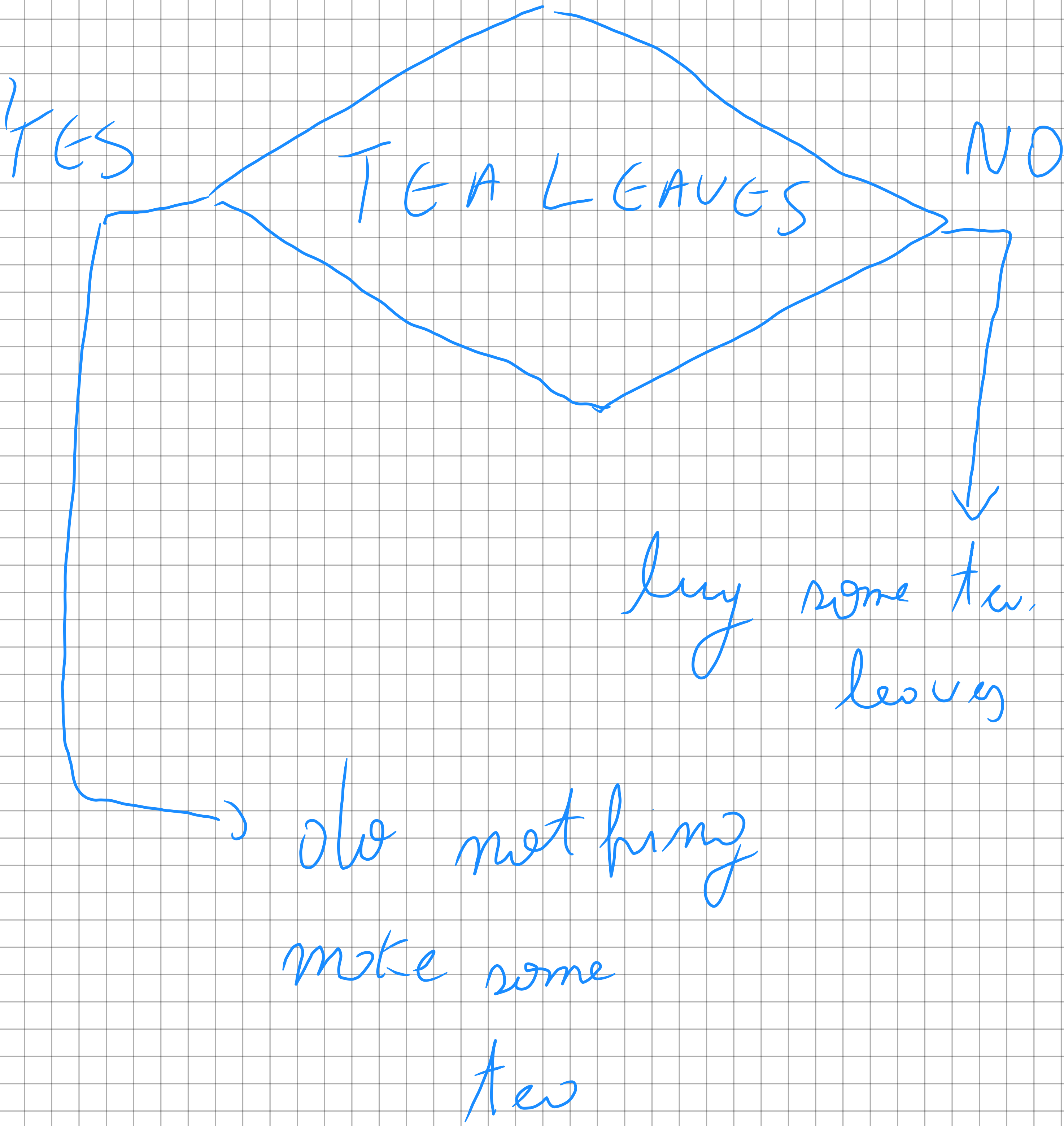
```
score = -85
has_passed_exam = True

if score >= 60:

    if has_passed_exam:
        #code
    else:
        #code

else:
    #code

#code after if Else
```


TEA LEAVES
YES
NO
buy some tea leaves
do nothing
make some tea

mini story

You're creating a notification system for a smart kettle.
It should remind the user only when the kettle has finished boiling

TASK:

- A variable kettle_boiled = TRUE
- If boiled, show "Kettle done! Time to make chai!"

```
kettle_boiled = TRUE

if kettle_boiled:
    print(Kelle Done! time to make chai")
```

Kelle Done! time to make chai

A local cafe wants a program that suggested a snack.

If a customer asks for cookies or samosa, it confirm the order

otherwise, it says it's not available

TASK

- Take a snack input
- if it's "cookies" or "samosa" confirm the order
- Else, show unavailability

```
snack = input("Enter your preffered snack: ")
print(f"User said: {snack}")

user said: BURGER
           -----

if snack == "cookies" or snack == "samosa":
    print(f"Great choice! We'll serve you {snack}")
else:
    print("Sorry, we only serves cookies or samosa with tea")
```

=) Enter your preferred snack

: samosa

⇓

Great choice! We'll serve you samosa

A tea stall offers different prices cup sizes

Write a program that calculates the prices based on size

TASK:

- input: "small", "medium", "large"
- Small → €10, Medium → €15, Large → €20
- If invalid: show "Unknown cup size"

```
input(" Choose your cup size
(small / medium / large ): ")
.lower

if cup == "small":
    print (" price is 10 Euros)
elif cup == "medium" :
    print (" price is 15 Euros)
elif cup == "large":
    print(" price is 20 Euros")
else :
    print (" Unknown cup size")
```

You're building a smart thermostat alert system:

- If the device_status is "active"
- And temperature > 35 → Warn High temperature
- Else: "Temperature normal
- If device is off → "Device is offline"

```
device_status = "active"

temperature = 38

if device_status == "active"
    if temperature > 35:
        print ("temperature ALERT")
else:
    print ("Device is offline")
```

You run an online tea store

If the order amount is more than 300€, delivery is free; otherwise, it cost 30€

TASK

- Input: order_amount
- Use ternary operator to decide delivery fee

```python
order_amount = int(input("Enter the order amount:"))

print(f"Order amount: {type(order_amount)}.")
```

```python
delivery_fees = 0
delivery_fees = 0 if order_amount > 300 else 30

print(f"Delivery fee is: {delivery_fees}.")
```

400 → 0 ; 100 → 30

You're building a ticket info system for railway app. Based on seat type, show its features

TASK:

- INPUT: "sleep", "AC", "luxury", "general";
- Match using match-case
- Unknown -> show: "invalid seat type"

```
input("Enter seat type
(sleeper/AC/general/luxury)".lower
()
seat_type
match seat_type:
    case "sleeper":
        print("Sleeper - No AC,
bed available")
    case "ac":
        print("AC - Air condit-
comfy ride")
    case: "general"
        print("General - Cheap option)
```

```
case: "luxury"
    print("Luxury - Premium
seats with meals")

case _:

    print("invalid seat
type")
```

CONDITIONALS ÎN PYTHON

IF (condi ie simpl)

Execut codul DOAR dac condi ia este adev rat (True)

```
x = 10

if x > 5:

   print("Mai mare decat 5")
```

Se execut pentru c 10 > 5

IF – ELSE

Dac condi ia nu este adev rat , execut alt bloc.

```
x = 3

if x > 5:

    print("Mai mare")

else:

    print("Mai mic sau egal")
```

Output:

Mai mic sau egal

IF – ELIF – ELSE

Pentru mai multe condi ii

```
x = 0

if x > 0:

    print("Pozitiv")

elif x < 0:

    print ("Negativ")

else:

    print("Zero")
```

Operatorii de compara ie

== → egal

!= → diferit

> → mai mare

< → mai mic

>= → mai mare sau egal

<= → mai mic sau egal

Operatorii logici

and -> ambele condiții TRUE

or -> cel puțin una TRUE

not -> inversează valoarea

```
x = 10

if x > 5 and x < 20:

    print("Intre 5 si 20")
```

Condi ie pe liste / dict

```
l = [1, 2, 3]

if 2 in l:

    print("Exista")
```

CONDITIONALS

- if
- if - else
- if - elif - else
- operatori comparatie
- operatori logici
- ternary operator

Ternary Operator (if scurt)

```
x = 10

mesaj = "Pozitiv" if x > 0 else "Negativ"

print(mesaj)
```

Diferen a: == vs =

= (assignment)

se folose te pentru atribuit valoare

x = 10 x prime te valoarea 10

== (compara ie)

verific dac dou valori sunt egale

x == 10

True dac x este 10

False altfel

Gre eal frecvent

```
if x = 10:   #    GRE IT
if x == 10:  #    corect
```

AND / OR (operatori logici)

and

se execut doar dac ambele sunt adev rate

ambele condi ii trebuie s fie True

```
if x > 0 and x < 20:

    print("Între 0  i 20")
```

or

cel pu in una trebuie s fie True

```
if x < 0 or x > 100:

  print("În afara intervalului")
```

se execut dac UNA e adev rat

Verificare existen   (list / dict)

În list (in)

```
l = [1, 2, 3]

if 2 in l:

    print("Exista")
```

True dac elementul e în list

În dic ionar (in)

```
d = {"a": 1, "b": 2}

if "a" in d:

    print("Cheie exista")
```

verific cheile, NU valorile

Condi ii multiple (elif)

pentru mai multe cazuri

if   prima condi ie

elif   alte condi ii

else   dac niciuna nu e adev rat

```
x = 0

if x > 0:

    print("Pozitiv")

elif x < 0:

    print("Negativ")

else:

    print("Zero")
```

```
x = 10

if x > 0 and x < 20:

    print("Între 0   i 20")

if x == 10:

    print("Este 10")

if 10 in [10, 20, 30]:

    print("Exista in lista")
```

= atribuire

== compara ie

and   toate True

or   cel pu in una True

in   verificare existen

elif   condi ii multiple

Exerci iul 1 – if / else

```
x = 8

if x > 10:

    print("Mai mare")

else:

    print("Mai mic sau egal")
```

Output:

Mai mic sau egal

pentru c   8 nu este > 10.

Exerci iul 2 – and (interval)

Condi ia:

```
if x >= 5 and x <= 20:

    print("În interval")
```

ambele trebuie True

verific : 5   x   20

Exerci iul 3 – in (list)

```
l = [1, 2, 3, 4]

if 3 in l:

    print("Exista")
```

Output:

Exista

Exerci iul 4 – if / elif / else

```
x = 0

if x > 0:

    print("Pozitiv")

elif x < 0:

    print("Negativ")

else:

    print("Zero")
```

Output:

Zero